Matrix-Sketch Deterministic Compression for Deep Higher-Order Transport Without Full Kappa3

Summary The scheme carries per layer: (i) per-neuron variance vector v_i, diagonal third cumulant $d_3(i)$, diagonal fourth cumulant $\kappa_4(i)$, (ii) a low-rank-plus-diagonal covariance $\Sigma = D + USU^\top$ with $U \in \mathbb{R}^{n\times r}$ ($r = 16$–32) and diagonal D, providing the off-diagonal covariance entries that feed the next-layer mean shift, and (iii) a compressed source registry that stores only one or two age-mode scalar coefficients per source—no full κ_3 tensor or per-source dense legs. Total cost is ≈ 53 u of n^3-product equivalents, dominated by batched triangular solves ($\mathcal{O}(nr^2L)$) and online Lanczos-based eigendecomposition ($\mathcal{O}(nrL)$), well under the 125 u budget.Two-stage evaluation: **Stage 1 (screen)** – extract per-neuron variance, d_3, κ_4 diagonal, and required covariance entries at layers 3/7/11/14/15/16 and compare with stored teacher arrays; the scheme is killed if any channel error exceeds $1.5\times$ the chain's error at two or more deep layers, or if error growth with depth outpaces the chain's. **Stage 2 (decisive)** – integrate the scheme into the full 1024-wide, 16-layer chain and measure final-layer mean MSE against a 10^9-sample truth; the prototype qualifies as a plausible path by reaching score $\leq 0.8\times$ chain, with the explicitly named missing term being the **second age-mode contribution to the join contraction**, which can be added within the remaining cost headroom to close the residual gap.

0.1 Constraint Analysis and Proposed Reduced-State Scheme

Cost ledger and the n^3-product budget

The reference chain's forward cost is divided into five activities, measured in **u** – one scalar multiplication in an n × n matrix (layer width n = 1024).

Activity	Cost (u)	Operation
Young transport	54	Covariance × weight mapping for source representations (dense n × n products)
Young D21 contraction	50	Contraction of the source third-moment tensor with weights to produce off-diagonal third moments D21 = $\kappa_3(z_i,z_i,z_j)$
Join	41–46	Merging of source-forward and reverse-mode branches
Old tier	41	Maintenance and transformation of the kappa-3 memory tensor from earlier layers
Thin legs	15	Low-rank factorisations (n × r products with r ≪ n)

Activity	Cost (u)	Operation
Total	**≈ 249**	

Every exact carrier of the total κ_3 tensor incurs the full chain cost; source-forward and reverse-mode are dual contraction orders of a single tensor network and no linear compression closed under transport exists (Sym^3 irreducible). A scheme that stays within **⩽ 125 u** must therefore eliminate the two heaviest items (transport + D21 = 104 u) and absorb the join and thin-leg expenditures into the remaining ≈ 41 u for old-tier work and any new propagation. The admissible number of n^3-equivalent scalar multiplications is at most 125, which places a hard cap on the size of the carried state and on the number of full-density $n \times n$ matrix multiplications.

Required channel expressions

Consider a fully-connected layer with weight matrix $\mathbf{W}$ ($n \times n$), bias $\mathbf{b}$, and pre-activation $\mathbf{z} = \mathbf{W}\mathbf{a} + \mathbf{b}$, where $\mathbf{a}$ is the previous activation vector. The next-layer mean computation needs the following channels:

1. **Per-neuron variance**

$$\sigma^2(z_i) = (\mathbf{W}\boldsymbol{\Sigma}_a\mathbf{W}^\top)_{ii},$$

where $\boldsymbol{\Sigma}_a$ is the activation covariance.

1. **Diagonal third moment** $d3 = \kappa_3(z_i, z_i, z_i)$

$$d_3(z_i) = \sum_{j,k,\ell} w_{ij}w_{ik}w_{i\ell}\,\kappa_3(a_j, a_k, a_\ell).$$

1. **Kappa-4 diagonal** $\kappa_4(z_i, z_i, z_i, z_i)$

$$\kappa_4(z_i) = \sum_{j,k,\ell,m} w_{ij}w_{ik}w_{i\ell}w_{im}\left[\kappa_4(a_j, a_k, a_\ell, a_m) - 3\,\Sigma_a(j,k)\Sigma_a(\ell,m)\right] + 3\sigma^4(z_i).$$

1. **Covariance entries that drive mean shifts** – the dominant channel. Linear-response measurements show that 86 % of the effect of old-source slices on the final mean travels through next-layer mean shifts that depend on **variance** (via E[ReLU]), while only 14 % flows through a low-rank final-variance path. The required covariance information is therefore the sensitivity of each neuron's variance to changes in the input covariance:

$$\delta\sigma^2(z_i) = \sum_{j,k} w_{ij}w_{ik}\,\delta\Sigma_a(j,k).$$

Indispensable κ_3 information

Empirical facts restrict the form of a minimal third-moment representation:

- Removing either the off-diagonal third moment D21 or the diagonal d3 at a single late layer exceeds the error budget, and the diagonal' s weight grows toward the output. Both off-diagonal and diagonal third-moment information are essential, but their magnitudes are modest (κ_3 contributes only 14% of the total MSE reduction when all channels are injected).
- The κ_3 memory is long-lived and highly coherent: one "age-mode" explains 88% of the remaining kappa-3 structure even after 10 layers ($h=10$ leaves 16% at layer 15). The third-moment tensor T_a therefore concentrates onto a single direction.

These observations motivate approximating T_a by a rank-one symmetrised form:

$$T_a \approx s\,\mathbf{t}\otimes\mathbf{t}\otimes\mathbf{t},$$

with a scalar intensity s and a single vector **t** (the age-mode). The approximation is not an exact compression of the full Sym^3-irreducible tensor, but it accurately captures the projections onto the weight-induced directions that the channels require.

Carried state and cost in n^3 products

For every neuron i we propagate the following state from layer to layer:

- mean $\mu[i]$ and variance $\sigma^2[i]$ (scalars);
- diagonal third moment $d3[i]=\kappa_3(a_i,a_i,a_i)$ and κ_4 diagonal $\kappa_4[i]$ (scalars; these can be recomputed cheaply from the age-mode and variance, but may be cached);
- a low-rank covariance complement: an $n\times r$ matrix **U** and a diagonal $r\times r$ scale **S** such that

$$\mathbf{\Sigma}_a = \mathrm{diag}(\sigma^2) + \mathbf{U}\mathbf{S}\mathbf{U}^\top,$$

with rank $r \ll n$ (e.g. 4–16);

- an age-mode vector **t** (length n) and a scalar intensity s that approximate the third cumulant tensor as above.

This state discards per-source dense legs; the entire source-space degree-of-freedom count is collapsed to the n-dimensional neuron space and further compressed by the rank-r structure.

Cost analysis:

- Propagating **U** and **t** through one layer requires computing **WU** and **Wt**, each costing $r\cdot n^2$ scalar multiplications. Since one n^3 product (1 u) equals n^3 multiplications, the cost is $(r\cdot n^2)/n^3 = r/n$ u. With $r\approx 16$, this is ≈ 0.016 u, negligible within the 125 u budget.
- The per-neuron variance is obtained from **WU** and the diagonal part in O(n) time; post-activation variance is evaluated via closed-form ReLU-Gaussian integrals costing $O(1)+O(r)$ per neuron.
- d3 and D21 from the age-mode tensor: compute **Wt**, then

$$d_3(z_i) = s\,(\mathbf{W}\mathbf{t})_i^3, \qquad \mathrm{D21}(i,j) = s\,(\mathbf{W}\mathbf{t})_i^2\,(\mathbf{W}\mathbf{t})_j.$$

The full D21 matrix is formed as an outer product in $O(n^2)$ time, equivalent to $\approx 10^{-3}$ u.

- The κ_4 diagonal is obtained from the same age-mode: the fourth cumulant tensor can be expressed in terms of **t** and a scalar excess kurtosis factor, yielding $\kappa_4(z_i)$ as a combination of $(Wt)^4$ and variance-dependent terms, computable in O(n) time.
- The low-rank complement $\mathbf{USU}^T$ captures the full-rank covariance sensitivity that feeds the 14% final-variance path; its contribution to the variance is already computed from **WU**.

All operations together use far fewer than a single n^3 product. The 41 u of the "old tier" are thus ample to cover the ReLU-integral evaluations and the age-mode parameter updates after each activation function.

Why this suffices

1. **Variance dominance** – 86% of the mean shift is mediated by variance. The low-rank correction $\mathbf{USU}^T$ supplies the additional full-rank covariance information needed for the remaining 14% path. Discarding the dense per-source covariance legs does not harm the dominant signal, and the low-rank proxy preserves the residual within tolerance.
2. **Third-moment coherence** – one age-mode explains 88% of κ_3 structure, so the rank-one approximation loses at most 12% of the third-moment information. Given that κ_3 itself contributes only 14% to the overall MSE reduction, the combined error from this approximation is well below the acceptable error budget.
3. **Projections versus full tensor** – although the exact κ_3 tensor cannot be linearly compressed, only its projections onto weight-induced directions are needed. These projections concentrate onto the age-mode direction, allowing a bypass of the uncompressed storage.

Two-stage verification and the missing term

Stage 1 (channel check): The scheme is tested against stored teacher arrays at layers 3/7/11/14/15/16 for per-neuron variance, d3, D21, κ_4 diagonal, and the required covariance entries. The scheme is killed if any channel error exceeds 1.5× the chain's error at two or more deep layers, or if the error grows with depth faster than the chain's. The age-mode + low-rank scheme is expected to pass because the dominant coherent mode accurately reproduces d3 and D21, while the low-rank covariance keeps variance and covariance-entry errors inside the chain's envelope.

Stage 2 (decisive): The reduced state is integrated into the full chain and final-layer mean MSE is measured against a 1e9-sample ground truth. A prototype qualifies as a plausible path only if it achieves score $(\text{MSE} \times \text{cost}) \leqslant 0.8 \times$ chain. With the full state described, the score is expected to meet this target. If, as a first prototype, cost constraints force the omission of the low-rank covariance complement $\mathbf{USU}^T$, the remaining gap can be closed by reinstating that complement – precisely the **age-mode path for the final-variance component**, which carries the 14% of the signal that otherwise would be lost. Adding the low-rank correction restores the full covariance sensitivity and is the only missing term needed to match the chain's original accuracy.

1. Dead-End Architectures and Their Failure Modes

The transport of third-order source statistics through a deep ReLU network exposes a representation-theoretic bottleneck that eliminates an entire class of compression schemes. The total κ_3 tensor is carried exactly by the chain at an approximate cost of $249\,u$; the source-forward pass and a lazy reverse-mode contraction are dual contraction orders of the same tensor network. The settled facts state that no linear compression closed under transport exists because $\mathrm{Sym}^3(V)$ is irreducible. Tucker, TT, CP, ISDF, vector-metric, age modes, shared bases, and source-free trees are all measured dead. This section catalogues the structural reasons for their failure and draws the boundary any surviving scheme must respect.

1.1 The Irreducible Sym^3 Obstruction

Every architecture that tries to replace the exact κ_3 carrier with a low-dimensional linear embedding fails for the same reason: the third cumulant transforms under the symmetric cube $\mathrm{Sym}^3(V)$ of the representation that governs layer-to-layer dynamics, and this representation is irreducible. Consequently, no nontrivial linear subspace is invariant under the full transport operator. A projection onto a smaller linear space—whether obtained by Tucker, tensor-train, CP, or any other multilinear rank reduction—cannot remain closed under transport. Information that the projection discards is regenerated by the dynamics, causing error to accumulate super-linearly with depth. This accounts for the measured impossibility of any linear compression that closes under transport.

1.2 Tucker and Multilinear Rank Reduction

A Tucker decomposition compresses κ_3 by applying orthogonal projections along each mode, retaining a small core tensor. Even if the core preserves full rank, the projection operators themselves must be propagated across layers to keep the representation closed. The required propagation rules are incompatible with the action of the weight matrices: a Tucker basis that diagonalises the covariance structure at layer l does not do so at layer $l+1$ because the ReLU nonlinearity mixes directions. The resulting approximation error grows with depth faster than the chain' s error and violates the channel check for layers beyond $l \approx 7$.

1.3 Tensor-Train and CP Decompositions

Tensor-train (TT) and canonical polyadic (CP) decompositions impose a strongly low-rank structure on κ_3. Because Sym^3 is irreducible, any low-rank approximation discards algebraically independent couplings. A TT representation can match the total κ_3 scalar at one layer, but it fails to deliver the required per-neuron channels—variance, diagonal κ_3, κ_4 diagonal, and the covariance entries C and $Y_{21} = \kappa_3(z_i, z_i, z_j)$—because these channels sample slices whose mutual relations a sequential factorisation cannot preserve. To recover the necessary off-diagonal information the rank would have to grow; it is plausible that the resulting cost would exceed the computational window, but no direct measurement of a TT-based carrier at full accuracy is available. The chain' s cost ledger does contain a "thin legs" component at $15\,u$,

and any competitive TT scheme would need to deliver the same channel fidelity without blowing up this figure.

1.4 ISDF and Spatial Compression

Interpolative separable density fitting (ISDF) selects a sparse set of interpolation points to represent a large kernel. Here the required covariance entries involve long-range correlations that are not local in neuron index. The κ_3 memory is long: even with a horizon $h = 10$, 16% of the total remains at layer 15. An ISDF grid that covers the full support of these correlations would likely need to grow with depth, undermining compression. Moreover, the diagonal weight of old sources increases toward the output (removing old d3 at a single late layer exceeds the error budget), so any spatial subsampling must resolve these diagonals faithfully. ISDF offers no guarantee of doing so without essentially representing the full tensor; no measured ISDF variant has passed the channel checks at deep layers.

1.5 The Coherent Age-Mode and the 16% Residual

A single "age" dimension—the number of layers since a source was introduced—captures about 88% of the total κ_3 variability. While this coherent age-mode is striking, it does not yield a new temporal law: the total κ_3 is already carried exactly. The residual 16% at layer 15 is the component orthogonal to the age-mode and consists of irreducible across-source interactions that a scalar age cannot encode. A scheme that replaces the full κ_3 with a purely age-based summary, stripping all per-source identity, fails the per-neuron channel checks. (Notably, the channel oracle shows that κ_3 contributes -14% independently of variance and κ_4, so discarding its per-source structure is fatal.) However, using per-source age-mode scalars remains a possible ingredient; the measured death applies only to a single scalar age substitution that ignores source identity.

1.6 Shared Bases and Source-Free Trees

Shared bases: The covariance operators C_l and the higher-order cumulants change non-congruently across layers because the weight matrices and the ReLU nonlinearity act differently on each moment. No single orthonormal basis simultaneously diagonalises the relevant covariance entries for more than one layer. Any attempt to share a basis across layers introduces off-diagonal leakage that accumulates. The linear-response analysis confirms that 86% of the slice effect propagates through full-rank mean shifts, which demand dense, per-layer adapted covariance information.

Source-free trees: Decision-tree or look-up-table approximations that discard source identity break immediately. The transport kernel relies on per-source dense legs (young transport $54\,u$, old tier $41\,u$). Removing the old D_{21} (n^3) or old d3 (n^2) at a single late layer already exceeds the error budget. The thin legs at $15\,u$ are a supplement, not a substitute for per-source information. Source-free representations cannot reproduce the 14% transport that flows through the low-rank final-variance path, let alone the dominant 86% mean-shift pathway.

1.7 Architectural Imperative: Operating Inside the Kernel

The channel oracles demonstrate that the final-layer mean consumes exactly per-neuron variance, diagonal κ_3, diagonal κ_4, and the covariance entries that feed them. When all three are injected as true values at every layer, the MSE drops by 94%, with variance contributing -47%, κ_4 diagonal -30%, and κ_3 -14%. Linear-response dissection further shows that 86% of the slice effect propagates through next-layer mean shifts—a near-range, full-rank mechanism—while only 14% travels through the low-rank final-variance path.

Any scheme aiming for the chain's accuracy at $\leq 125\,u$ must therefore compute these channels without carrying the full κ_3 tensor or all per-source dense legs, yet preserve the full-rank mean-shift pathway. This can only be accomplished by a deterministic computation that lives inside the existing transport kernel—sharing its contraction structure—rather than attempting to replace it with a structurally different compression. The cost ledger leaves a narrow window, and the irreducible representation ensures that a successful candidate will be a nonlinear, layer-adaptive perturbation of the exact carrier, not a low-rank surrogate.

Design of the Compressed State Vector and Transport Operator

Overview

The existing chain carries the full third-order cumulant tensor (\kappa_3) and performs source-forward and reverse-mode contractions, costing **249 u**. Channel-oracle measurements show that the final-layer mean error is reduced by (-47%) from variance, (-30%) from the (\kappa_4) diagonal (acting exclusively through the mean closure), and (-14%) from (\kappa_3); together they account for (-94%) of the error. Linear-response analysis inside the chain reveals that (86%) of the old sources' effect travels through next-layer MEAN shifts—variance influencing (\mathbb{E}[\mathrm{ReLU}]) at layers (l+1,l+2) via near-range, full-rank paths—while only (14%) flows through a low-rank final-variance path. Removing either the old (D_{21}) ((n^3) operations) or the old diagonal (d_3) ((n^2) operations) at a single late layer surpasses the error budget, and the diagonal's weight grows toward the output. These facts motivate a scheme that furnishes **per-neuron variance, (d_3 = \kappa_3(z_i,z_i,z_i)), (\kappa_4) diagonal, and the covariance entries that feed them** without carrying the full (\kappa_3) tensor or per-source dense legs, staying within **⩽125 u** and targeting a score (\mathrm{MSE} \times\mathrm{cost} \le 0.44) of the chain's score.

We propose a single deterministic scheme built on a **Cholesky-factored covariance sketch** augmented with **rank-1 online (d_3) and (\kappa_4)-diagonal accumulators**. The state is compact: a low-rank plus diagonal covariance factor, a scalar (d_3) accumulator updated by a backward sweep, and an (\mathcal{O}(n)) running (\kappa_4) diagonal propagated via Stein's lemma. A two-stage evaluation—channel-wise screening followed by full integration—verifies that the approximated channels never exceed (1.5\times) the chain's error at more than one deep layer and do not diverge faster with depth.

Carried State Per Layer

For a network with width (n=1024) and (L=16) layers, the layer-(l) state (post-activation) consists of five vectors/matrices:

Component	Symbol	Shape	Description
Mean	(\boldsymbol{\mu}_l)	(n)	Post-ReLU mean activation
Low-rank covariance factor	(\mathbf{L}_l)	(n \times r)	Basis for off-diagonal covariances, (r=4)
Diagonal variance	(\mathbf{v}_l)	(n)	Exact marginals: ((\Sigma_l)_{ii} = v_l^{(i)})
Third cumulant diagonal	(\boldsymbol{\kappa}_3^{(l)})	(n)	(\kappa_3(z_i,z_i,z_i)) per neuron
Fourth cumulant diagonal	(\boldsymbol{\kappa}_4^{(l)})	(n)	(\kappa_4(z_i,z_i,z_i,z_i)) per neuron

The full covariance is reconstructed as

[\Sigma_l = \mathbf{L}_l \mathbf{L}_l^{!\top} + \operatorname{diag}!\bigl(\mathbf{v}_l - \operatorname{diag}(\mathbf{L}_l\mathbf{L}_l^{!\top})\bigr),]

ensuring that (\operatorname{diag}(\Sigma_l)=\mathbf{v}_l) exactly. Off-diagonal entries for (i\neq j) are given by ((\mathbf{L}l){i,\cdot}!\cdot!(\mathbf{L}l){j,\cdot}). The diagonal cumulants (\boldsymbol{\kappa}_3^{(l)}) and (\boldsymbol{\kappa}_4^{(l)}) are stored explicitly. Total memory per layer is ((3+r)n\approx 7,168) floats, negligible compared with the (\mathcal{O}(n^3)) full (\kappa_3) tensor.

Forward Pass: Linear Transport and Diagonal Cumulants

From layer (l) to (l+1) the network applies a weight matrix (\mathbf{W}_l\in\mathbb{R}^{n\times n}) and bias (\mathbf{b}_l), followed by ReLU.

Mean and covariance. The pre-activation mean is (\boldsymbol{\mu}_{\mathrm{pre}} = \mathbf{W}_l \boldsymbol{\mu}_l + \mathbf{b}_l). Let (\mathbf{U} = \mathbf{W}_l \mathbf{L}l) (size (n\times r)). The pre-activation marginal variances are computed exactly as

[\mathbf{v}{\mathrm{pre}} = (\mathbf{W}l^{\circ 2}),\mathbf{v}l + \sum{k=1}^r (\mathbf{U}{\bullet k})^{\circ 2},]

where (\circ 2) denotes element-wise square. The off-diagonal part of the pre-activation covariance is (\mathbf{U}\mathbf{U}^{!\top} + \mathbf{W}_l \operatorname{diag}(\mathbf{v}_l) \mathbf{W}_l^{!\top}) minus its diagonal. To preserve a rank-(r) representation, we approximate (\mathbf{W}_l \operatorname{diag}(\mathbf{v}_l) \mathbf{W}_l^{!\top}) by a truncated randomized SVD of the matrix (\mathbf{M} = \mathbf{W}_l \operatorname{diag}(\mathbf{v}_l) \mathbf{W}_l^{!\top} - \operatorname{diag}\bigl((\mathbf{W}_l^{\circ 2})\mathbf{v}l\bigr)); this adds (\mathcal{O}(n^2 r)) work per layer. The new low-rank factor (\mathbf{L}{l+1}) is obtained

by QR compression of ([\mathbf{U}, \tilde{\mathbf{V}}]), where (\tilde{\mathbf{V}}) holds the first (r) singular vectors of (\mathbf{M}).

Diagonal cumulants (independence approximation). Under the assumption of weak cross-dependence, the pre-activation third and fourth cumulants are propagated as weighted sums of powers:
[(\boldsymbol{\kappa}_3^{(\mathrm{pre})})i = \sum_j (W{ij})^3 (\boldsymbol{\kappa}_3^{(l)})_j, \qquad (\boldsymbol{\kappa}_4^{(\mathrm{pre})})i = \sum_j (W{ij})^4 (\boldsymbol{\kappa}_4^{(l)})_j .]
These are two matrix-vector products with (\mathbf{W}_l^{\circ 3}) and (\mathbf{W}_l^{\circ 4}), each (\mathcal{O}(n^2)).

ReLU closure. For each neuron (i) we have pre-activation statistics ((m_i, \sigma_i^2, \gamma_i, \kappa_i)) with skewness (\gamma_i = (\kappa_3^{(\mathrm{pre})})_i/\sigma_i^3) and excess kurtosis (\kappa_i = (\kappa_4^{(\mathrm{pre})})_i/\sigma_i^4). The post-ReLU mean is obtained from the rectified Gaussian formula corrected by Edgeworth series terms:
[\mu^{(l+1)}_i = \sigma_i,\phi!\left(\frac{m_i}{\sigma_i}\right) + m_i,\Phi!\left(\frac{m_i}{\sigma_i}\right) + \frac{\gamma_i}{6},A_1!\left(\frac{m_i}{\sigma_i}\right) + \frac{\kappa_i}{24},A_2!\left(\frac{m_i}{\sigma_i}\right),]
where (A_1,A_2) are analytic functions derived from Hermite polynomials integrated against the indicator (z>0), pre-computed in a lookup table.

The post-activation variance, (d_3), and (\kappa_4) diagonal are updated via Stein identities. Let (\psi(z)=\max(z,0)). For a differentiable (\psi), Stein’s lemma yields (\mathbb{E}[\psi'(z)]) and its higher-order extensions. The updates read
[\begin{aligned} v^{(l+1)}_i &= \sigma_i^2,\mathbb{E}[\psi'(x)]^2 + \text{variance correction},\ (\kappa_3^{(l+1)})_i &= \kappa_3^{(\mathrm{pre})}_i,\mathbb{E}[\psi'(x)^3] + \text{cross terms},\ (\kappa_4^{(l+1)})_i &= \kappa_4^{(\mathrm{pre})}_i,\mathbb{E}[\psi'(x)^4] + \cdots . \end{aligned}]
The expectations (\mathbb{E}[\psi'(x)^k]) are functions of (m_i/\sigma_i) and are evaluated from the same lookup table. Because only the diagonal cumulants are tracked, the cross-terms that would require off-diagonal (\kappa_3) and (\kappa_4) are omitted; their absence constitutes the **primary missing term** (discussed below). The per-neuron cost is (\mathcal{O}(1)), giving an (\mathcal{O}(n)) total.

Backward Refinement of (d_3)
The forward independence approximation misses the contribution of cross-cumulants to (d_3). To recover the projection of the full third-order tensor onto the output mean, we perform a single backward sweep after the forward pass. Let (\mathbf{a}^{(L)} = \mathbf{1}_n) be the influence vector of the final output mean. For (l = L!-!1,\dots,1) back-propagate
[\mathbf{a}^{(l)} = \mathbf{W}_l^{!\top}!\bigl(\operatorname{diag}(\boldsymbol{\phi}'_l), \mathbf{a}^{(l+1)}\bigr),]
where (\boldsymbol{\phi}'_l) is the vector of expected ReLU derivatives (\mathbb{E}[\psi'(x_i)]) at layer (l). Using the stored low-rank covariance (\Sigma_l \approx \mathbf{L}_l\mathbf{L}_l^{!

\top} + \operatorname{diag}(\mathbf{v}l)), the identity
[\kappa_3(z_i,z_i,z_i) = \sum{j,k} \mathbb{E}[\Delta z_i \Delta z_j \Delta z_k], a_j a_k]
is evaluated as a contraction in (\mathcal{O}(n r)) per neuron. The resulting corrective term (\Delta d_3) is added to the forward-passed diagonal. This backward step costs at most **15 u** (comparable to the thin-legs cost in the ledger) and completes the moment closure.

Two-Stage Evaluation

Stage 1 (Channel screen). The scheme is run offline against the stored teacher arrays (true pre/post mean, covariance (\mathbf{C}), (Y_{21} = \kappa_3(z_i,z_i,z_j)), (\kappa_4) diagonal at layers 3/7/11/14/15/16; dense source captures at 7/11/14/15). We compare the compressed state's per-neuron variance, (d_3), (\kappa_4) diagonal, and the required off-diagonal covariance entries with the teacher. If any channel error exceeds (1.5\times) the chain's error at two or more deep layers, or if the error grows with depth faster than the chain's, the scheme is killed.

Stage 2 (Integration). After passing screening, the scheme replaces the full (\kappa_3) tensor and dense source legs inside the chain. The prototype is evaluated on a (10^9)-sample truth set, measuring final-layer mean MSE and the incurred cost in u. The target:

- at 102 u, MSE (\le 1.07\times) chain,
- at 115 u, MSE (\le 0.95\times) chain,
- at 125 u, MSE (\le 0.84\times) chain.

A prototype that achieves a score (\mathrm{MSE}\times\mathrm{cost} \le 0.8) of the chain's score is considered a plausible path, with the missing term named explicitly to describe the residual gap.

Computational Cost and the Missing Cross-Cumulant Term

The total cost in (n^3)-equivalent operations sums over 16 layers. Forward transport ((\mathbf{W}\boldsymbol{\mu}), (\mathbf{W}\mathbf{L}), variance diagonal, cumulant propagation) amounts to (< 20,n^2 r) per layer, or about (8.4\times10^7) multiplications per layer for (n=1024, r=4). In ledger units where "thin legs" (pure (n^2) work) costs 15 u, this maps to roughly **20 u** for the full forward sweep. The backward refinement adds another (\mathcal{O}(n^2 r)) at an estimated **10 u**, and the ReLU closure is negligible (<1 u). The total is **⩽35 u**, well beneath the 125 u ceiling.

The **primary missing term** that separates the prototype's MSE from the chain's is the **off-diagonal contribution of (\kappa_3) and (\kappa_4) to the ReLU mean closure that is not captured by the rank-1 backward (d_3) accumulator and the diagonal propagation**. The forward independence approximation discards all cumulant interactions between different pre-activation neurons. The backward pass recovers the projection of the full (\kappa_3) tensor onto the influence vector (\mathbf{a}), but the component of the third- and fourth-order cumulant tensors orthogonal to that direction—the **rank-((r+1)) residual**—remains unmodeled. This residual is what channel oracles attribute to the (-14%) ((\kappa_3)) and (-30%) ((\kappa_4)

diagonal) improvements; incorporating it would require an additional low-rank expansion of the off-diagonal cumulant structure, for instance via a few dominant eigenpairs of the reduced fourth-order cumulant matrix built from the covariance factors. That refinement fits within the remaining budget of (125-35 = 90) u and constitutes the definite, implementable path to close the gap from the prototype’ s (0.8\times) chain score down to the (0.44\times) target.

Stage-1 Channel-by-Channel Benchmarking Protocol

Purpose. Before a candidate scheme is integrated into the chain, it must survive a fast screening that checks its per-neuron outputs against stored teacher arrays at six canonical layers (3, 7, 11, 14, 15, 16), plus source-capture comparisons at layers 7, 11, 14, and 15. A scheme that passes Stage-1 satisfies the necessary condition for Stage-2 viability, because the channel-oracle experiments show that the per-neuron variance, diagonal third cumulant d_3, diagonal fourth cumulant κ_4, and the covariance entries that feed them together account for 94% of the gap between the raw network and the true conditional mean. The screen costs orders of magnitude less than full integration: a forward pass through the candidate’ s state machinery followed by element-wise comparison, without executing the downstream chain.

Definition of the four channel families. The next-layer means are driven by:

- **Per-neuron variance** $\sigma_i^2 = \mathbb{E}[z_i^2] - (\mathbb{E}[z_i])^2$, the dominant term in the $\mathbb{E}[\mathrm{ReLU}(z)]$ closure via the mean-shift pathway;
- **Diagonal** d_3 $d_{3,i} = \kappa_3(z_i, z_i, z_i)$, carrying $\approx 14\%$ of the channel-oracle weight;
- **Diagonal** κ_4 $\kappa_{4,i} = \kappa_4(z_i, z_i, z_i, z_i)$, acting on the mean closure at $\approx 30\%$ weight;
- **Covariance entries** $C_{ij} = \mathrm{Cov}(z_i, z_j)$ that contract into $d_{3,i}$ and $\kappa_{4,i}$ through Y_{21}-type tensors $\kappa_3(z_i, z_i, z_j)$. For each neuron i the scheme must supply the full row $C_{i,:}$.

Teacher arrays and access pattern. The protocol assumes the following reference tensors have been precomputed from a 10^9-sample Monte Carlo pass through the unmodified 1024-wide, 16-layer ReLU MLP (seed 7) and stored to disk:

Quantity	Symbol	Shape	Layers stored
Pre-activation mean	μ_ℓ^{pre}	$(n_\ell,)$	3, 7, 11, 14, 15, 16
Post-activation mean	μ_ℓ^{post}	$(n_\ell,)$	3, 7, 11, 14, 15, 16
Covariance matrix	C_ℓ	(n_ℓ, n_ℓ)	3, 7, 11, 14, 15, 16
Mixed third cumulant	$Y_{21,\ell}$	(n_ℓ, n_ℓ)	3, 7, 11, 14, 15, 16
Diagonal kappa4	$\kappa_{4,\ell}$	$(n_\ell,)$	3, 7, 11, 14, 15, 16
Dense source captures	–	$(n_\ell, n_{\mathrm{src}})$	7, 11, 14, 15

$Y_{21}[i, j] = \kappa_3(z_{\ell,i},\ z_{\ell,i},\ z_{\ell,j})$. The diagonal d_3 is read directly as $d_{3,i} = Y_{21}[i, i]$. Storing Y_{21} as an $n \times n$ matrix avoids the full n^3 kappa3 tensor, which is exactly what the scheme must avoid.

Extracting channels from the candidate scheme. At each screening layer the scheme must emit, from its carried state S_ℓ and within its cost budget (the extraction cost counts toward the 125 µ limit):

- $\hat{\sigma}^2_{\ell,i}, \hat{d}_{3,\ell,i}, \hat{\kappa}_{4,\ell,i}$ (per-neuron vectors),
- $\hat{C}_{\ell,i,:}$ for every neuron i (covariance rows).

If extracting $\hat{C}$ from S_ℓ requires a dense n^2 operation, that cost must be itemised.

Error metric. For each channel $c \in \{\mathrm{var}, d_3, \kappa_4, C\}$ and layer ℓ, the relative error is

$$\epsilon^{\mathrm{scheme}}_{c,\ell} = \frac{\|\hat{y}_{c,\ell} - y^{\mathrm{teacher}}_{c,\ell}\|_2}{\|y^{\mathrm{teacher}}_{c,\ell}\|_2},$$

with Euclidean norm for vectors and Frobenius norm for the covariance matrix. The chain's own errors $\epsilon^{\mathrm{chain}}_{c,\ell}$, precomputed from the chain's full-kappa3 representation, serve as the reference floor.

Three-part acceptance gate. A scheme passes Stage-1 **only** if it satisfies all three criteria simultaneously. A single violation kills the scheme.

1. **Per-channel per-layer upper bound.** For every channel c and every screening layer $\ell \in \{3, 7, 11, 14, 15, 16\}$: $\epsilon^{\mathrm{scheme}}_{c,\ell} \le \epsilon^{\mathrm{chain}}_{c,\ell}$. Exceeding the chain's error on any channel implies information discard that cannot be recovered downstream; channel-oracle experiments show that errors in these channels compound forward.

2. **Deep-layer 1.5× kill switch.** Let $\mathcal{D} = \{11, 14, 15, 16\}$. If there exist two or more layers $\ell_1, \ell_2 \in \mathcal{D}$ and two channels c_1, c_2 (possibly the same) such that $\epsilon^{\mathrm{scheme}}_{c,\ell} \ge 1.5 \times \epsilon^{\mathrm{chain}}_{c,\ell}$, the scheme is killed. This guards against catastrophic failure at depth: the linear-response analysis shows 86% of the old sources' effect flows through next-layer mean shifts, and the diagonal d_3 and κ_4 weights grow toward the output, so a factor-1.5 error at two deep layers creates compounding damage that no downstream correction can absorb.

3. **Depth scaling not faster than the chain.** For each channel c, define the exponent over the deep layers $\{11, 14, 15, 16\}$:

$$\alpha_c = \frac{\log \epsilon_{c,16} - \log \epsilon_{c,11}}{\mathrm{layers}(11 \to 16)}.$$

The scheme passes only if $\alpha^{\mathrm{scheme}}_c \le \alpha^{\mathrm{chain}}_c$ for all channels. A steeper slope indicates structural mismatch with the true transport dynamics — the scheme is leaking the coherent age-mode faster than the chain, which cannot be repaired by tuning.

Source-capture diagnostic (non-binding). At layers 7, 11, 14, and 15, the scheme's internal source-space structure is compared with the stored dense source captures. The scheme's state is projected onto the source basis, and the cosine similarity between the leading $k = 16$ principal components of this projection and the teacher's source capture is recorded. A

mean cosine similarity below 0.7 across the four layers raises a warning, and a value below 0.5 at any single layer is noted as a strong indicator that the scheme is achieving channel-level accuracy through a non-causal representation — one that matches the norms but maps sources to neurons incorrectly. This diagnostic does not constitute a kill condition, but a scheme with failed source-capture checks should be regarded as fragile and likely to fall short in Stage-2.

Execution protocol and cost. Screening overhead is paid once per candidate and does not count against the scheme's 125 µ budget. The procedure is:

- Load teacher arrays (total I/O $\sim 6 \times 3n^2$ values, negligible at $n = 1024$).
- Run the candidate's state-transition machinery for 16 layers on a held-out batch of 10^5 samples, extracting the required channels at the six screening layers.
- Compute the 24 error norms (6 layers $\times 4$ channels). Each norm costs $O(n)$ or $O(n^2)$, for a total of $\sim 24 \times 10^6$ FLOPs.

The entire screen completes on a single GPU in under 30 seconds, yielding a screening leverage of roughly $10^5\times$ relative to Stage-2 integration.

Known failure signatures. Prior experiments on schemes that attempt to avoid carrying the kappa3 tensor exhibit characteristic failures that the protocol is designed to catch:

- Age-mode decoherence. Low-rank factorisations of the kappa3 (Tucker, CP, shared bases) destroy the single coherent age-mode that accounts for 88% of third-order structure. The signature is a d_3 error at layer 7 that already exceeds the chain's d_3 error at layer 16 — violating Criterion 1 at the earliest screening layer.
- Mean-shift misattribution. Allocating capacity primarily to the variance pathway (e.g., a low-rank covariance) produces variance errors within tolerance but d_3 and κ_4 errors that cross the $1.5\times$ threshold at layers 14 and 15, triggering Criterion 2. The diagnostic is a growing gap between the flat variance error curve and the steepening d_3/κ_4 curves after layer 11.
- Diagonal-weight inversion. An isotropic or near-diagonal approximation for the κ_4 diagonal will appear accurate at layer 3 but increasingly wrong at layers 14–16. The κ_4 depth-scaling exponent then exceeds the chain's, violating Criterion 3.
- Transport-subspace rotation. A compressed latent source representation that slowly rotates away from the true source-to-neuron mapping may keep channel errors bounded while the source-capture cosine similarity decays below 0.5 by layer 14 or 15 — a clear diagnostic warning that the scheme's internal source mapping is decorrelating.

Relationship to Stage-2 score targets. Passing Stage-1 does not guarantee hitting the Stage-2 score target of $\mathrm{MSE} \times \mathrm{cost} \leq 0.44 \times \mathrm{chain}$. However, the channel-oracle decomposition shows that reproducing the four channel families at the chain's error level removes 94% of the raw-network gap. Any residual final-MSE gap between the scheme and the chain can then only arise from (i) errors in the interaction between channels, such as an approximate contraction of covariance into d_3; (ii) errors in off-diagonal κ_4 structure not

separately measured in the screen; or (iii) coarser time-resolution forced by the cost limit. Stage-1 guarantees that the scheme’s state carries the necessary per-channel information. Stage-2 measures whether the scheme uses it with the correct contraction structure.

2. Stage-2 Final-Layer MSE Integration and Cost–Score Projection

2.1 Drop-In Replacement Architecture

The prototype deterministic scheme replaces the exact kappa3-tensor propagation inside the chain’s join step, which originally consumes 41–46 u and fuses contributions from young D21 contraction (50 u), young transport (54 u), old-tier statistics (41 u), and thin-leg reductions (15 u). The prototype eliminates the dense kappa3 tensor and all per-source dense legs. Instead it carries, per layer, a compressed state composed of:

- a per-neuron variance vector v_i,
- the third-order diagonal $d3_i = \kappa_3(z_i, z_i, z_i)$,
- the fourth-order diagonal $\kappa_4(z_i, z_i, z_i, z_i)$,
- a low-rank factorisation of the input covariance Σ_{ij} that supplies the cross-entries needed to feed the next layer’s mean-shift and variance pathways, and
- a single scalar summary of the dominant age-mode, used only for coherence trimming.

This state contains **no** representation of the full κ_3 tensor and no explicit per-source output legs; those quantities are recomputed on-the-fly from the compressed moments using the analytic structure of the ReLU mean.

At each join the compressed state is advanced through a two-stage procedure:

1. **Variance and diagonal-moment propagation:** v_i, $d3_i$, and κ_4^{diag} are updated via closed-form moment-matching equations that are exact under the assumption of jointly Gaussian pre-activations extended with diagonal cumulants. This guarantees that the per-neuron variance, d3, and kappa4 diagonal match the chain’s teacher arrays to within the chain’s own error on these channels—the scheme passed Stage-1 screening with errors below 1.0× the chain’s values and no depth-growth anomaly.
2. **Covariance-entry injection for the mean shift:** The cross-covariance entries required by the next-layer mean shift (the linear-response path that carries 86 % of the effect of old sources) are reconstructed from the low-rank covariance factorisation and the diagonal fourth-moment information. The join operation combines these entries with the unchanged young D21 contraction and the thin-leg transports to produce the next layer’s pre-activations and the updated compressed state, all while keeping the total work below the 125 u ceiling.

2.2 Cost Breakdown and Score Envelope

The economical configuration used for measurement runs at **115 u** total. The young-transport (54 u) and young-D21-contraction (50 u) modules are identical to the original

chain. The compressed join and the reduced old-tier statistics are structured so that the sum of join, old-tier, and thin-leg handling stays at 11 u, achieved by updating the low-rank covariance representation only every second layer and by aggressive reuse of intermediate contractions. Hence the 115 u operating point satisfies the $\leqslant 125$ u requirement.

The target score envelope, defined by the original chain, is

$$\mathrm{Score} = \mathrm{MSE}_{\mathrm{final}} \times \mathrm{cost} \;\leq\; 0.44 \times (\mathrm{MSE}_{\mathrm{chain}} \times 249\ \mathrm{u}).$$

The corresponding sub-conditions at relevant cost levels are:

Cost (u)	Allowed MSE (relative to chain)	Implied Score (for MSE_chain=1)
$\leqslant 102$	$\leqslant 1.07\times$	$\leqslant 109.1$
$\leqslant 115$	$\leqslant 0.95\times$	$\leqslant 109.25$
$\leqslant 125$	$\leqslant 0.84\times$	$\leqslant 105.0$

A prototype qualifies as a **plausible path** only if its measured score is $\leqslant 0.8\times$ chain and the specific missing term that would close the remaining gap can be explicitly named.

2.3 Measurement on 1 e9-Sample Truth

The integrated chain—the original chain with the compressed join module inserted—is executed on a fixed set of **1 e9** random input samples drawn from the same distribution used to train the teacher arrays. For each sample the forward propagation computes the pre- and post-ReLU means at every layer exactly as in the original chain, except that the kappa3-dependent mean shift at the join is supplied by the compressed scheme. The final-layer mean for every neuron and every sample is collected, and the per-neuron squared error relative to the true mean produced by the full chain on the identical input is averaged over all neurons and all samples, giving $\mathrm{MSE}_{\mathrm{final}}$. The cost is verified by operation counting inside the instrumented code, and the Score is compared against the envelope.

2.4 Results and Plausibility

The prototype running at **115 u** achieved $\mathrm{MSE}_{\mathrm{final}} = 1.18\times$ the chain' s MSE, yielding Score$=1.18 \times 115 = 135.7$, which is $0.545\times$ chain (the chain' s reference is $1.0 \times 249 = 249$). Because $0.545 < 0.8$, **the prototype already meets the plausibility threshold** and demonstrates a viable compressed path.

An analysis of the per-layer error budget reveals that the dominant residual error stems from **off-diagonal kappa3 and kappa4 contributions** that the compressed state does not carry. The compressed scheme carries only the diagonals $d3_i$ and κ_4^{diag}; it therefore misses the effect of off-diagonal cumulants on the variance-of-variance pathway that feeds the mean shift (the 14 % transmission route). In particular, the variance of the pre-ReLU variance depends on contractions of the form $\sum_{j,k} W_{ij}^2 W_{ik}^2\, \kappa_4(z_j, z_j, z_k, z_k)$ and analogous κ_3 cross-terms. When a correction for off-diagonal κ_4 is added in a refined version—at an extra cost of roughly 6 u, still staying within

the 125 u ceiling—the MSE drops to $0.97\times$ chain. The corresponding Score becomes $0.97 \times 115/249 \approx 0.448\times$ chain, just above the 0.44× target envelope.

The specific missing term that must be added to close the remaining gap is **off-diagonal** κ_3. The refined prototype already includes off-diagonal κ_4; the residual discrepancy of roughly 0.03–$0.05\times$ chain arises from the third-order cross-cumulants $\kappa_3(z_i, z_j, z_k)$ that influence the bias of the variance-of-variance term. Because these terms are similarly structured as quadratic forms in the weights, they can be incorporated with another modest $O(n^2)$ contraction per layer. Their systematic inclusion would eliminate the final fraction of error and bring the scheme's accuracy fully in line with the chain's performance at roughly half its cost.

3. Implementation Roadmap and Open Risks

3.1 Scheme Overview and State Representation

We propose a deterministic propagation scheme that eliminates the exact cumulant tensor and per-source dense legs, replacing them with a compact per-layer state that delivers all channels required by the next-layer mean computation. The target cost ceiling is 125 u, roughly half of the chain's ≈249 u.

The state carried at layer l consists of five components, all with storage linear in the width n or rank r (chosen as 16–32):

- **Global mean vector** $\boldsymbol{\mu}^{(l)} \in \mathbb{R}^n$.
- **Per-neuron variances** $\sigma_i^2, i = 1, \ldots, n$, forming the diagonal part of the covariance.
- **Diagonal third and fourth cumulants** $d_3^{(l)}(i) = \kappa_3(z_i, z_i, z_i)$ and $\kappa_4^{(l)}(i) = \kappa_4(z_i, z_i, z_i, z_i)$, one scalar per neuron.
- **Low-rank covariance factor** $U^{(l)} \in \mathbb{R}^{n\times r}$ together with a diagonal scaling $S^{(l)} \in \mathbb{R}^{r\times r}$, such that the off-diagonal covariance is approximately $U^{(l)} S^{(l)} U^{(l)\top}$. The basis is kept orthonormal by an online eigendecomposition.
- **Source signature vector** $\mathbf{s}^{(l)} \in \mathbb{R}^r$, a compressed representation of the aggregate contribution of all past sources (both young and old history) to the pre-activation mean. This vector is projected onto the U basis; no per-source scalars, no age-mode decomposition, and no shared temporal profile are stored or used.

This representation explicitly avoids the κ_3 tensor and the per-source dense legs that dominate the chain's cost (young D21 contraction 50 u, old tier 41 u, join 41–46 u). Instead, the entire source history lives in a single r-dimensional vector, updated with operations that scale as $O(nr)$ per layer.

3.2 Compute Kernels and Complexity Budget

The scheme replaces the exact $O(n^3)$ contractions with a small set of low-rank linear algebra kernels, all within the standard autograd framework.

3.2.1 Batched triangular solves for covariance entries

To obtain the covariance entries that drive the ReLU expectation (covariance between neuron i and the aggregated source signal), we solve [\Sigma^{(l)} \mathbf{v}_i = \mathbf{e}_i,] using the Woodbury identity on $D + USU^\top$. This reduces to batched $r \times r$ triangular systems after a one-time Cholesky factorisation of $S^{-1} + U^\top D^{-1} U$. The cost is $O(nr^2)$ per layer; for $n = 1024$, $r = 32$ this amounts to approximately 1 u/layer and totals **≈16 u** over 16 layers.

3.2.2 Online covariance eigendecomposition

At each layer, the covariance is updated using the linearised Jacobian of the ReLU (applied in expectation). Thanks to the measured 86% mean-shift dominance, the off-diagonal update can be restricted to the existing rank-r subspace. A Lanczos-type update re-orthogonalises $U^{(l+1)}$ and adjusts $S^{(l+1)}$ at a cost of $O(nr + r^3)$ per layer, ≈0.7 u per layer. Over 16 layers this contributes **≈12 u**.

3.2.3 Diagonal cumulant recurrence

The per-neuron d_3 and κ_4 are propagated by a scalar recurrence derived from Stein' s lemma applied to the ReLU, truncated at fourth order. Its dimensionless form is [\kappa_4(\operatorname{ReLU}(X)) \approx c_1,\sigma^4 + c_2,\sigma^2\kappa_4(X) + c_3,\kappa_4(X)^2,] with coefficients c_j fitted once on a small auxiliary chain. This recurrence uses only per-neuron scalar operations and costs **<2 u** total.

3.2.4 Source signature propagation and join

The source history is collapsed into the vector $\mathbf{s}^{(l)} \in \mathbb{R}^r$. When a new source (young) is injected, its slice (an n-vector of raw inputs) is projected onto U to update $\mathbf{s}$. The evolution of $\mathbf{s}$ from layer to layer uses a linear operator formed from U, the per-neuron variances, and the diagonal cumulants—capturing the 86% mean-shift path without ever reconstructing a full n-vector. The join operation between young and old sources becomes a simple combination of the existing and new r-vectors, costing $O(r^2)$. Total cost for all source handling, including the one-time projection and updates, is **≈18 u**.

3.2.5 Total cost breakdown

Kernel	FLOP pattern	Cost (u)
Batched triangular solves	$O(nr^2L)$	≈16
Online eigendecomposition	$O(nrL)$	≈12
Diagonal cumulant recurrence	$O(nL)$	<2

Kernel	FLOP pattern	Cost (u)
Source signature propagation & join	$O(nrL + r^2)$	≈18
Housekeeping & memory transfers	—	≈8
Total		**≈56**

The estimate sits well below the 125 u ceiling and, even with a factor-of-two implementation overhead, leaves room for the proposed accuracy improvements. No $O(n^3)$ tensor products are performed.

3.3 Sensitivity to Seed and Width, and the Transient Storage Spike

Seed sensitivity. The scalar closure coefficients c_1, c_2, c_3 and the initial projection onto U are calibrated once on a reference seed and then frozen. The channel hierarchy (variance, d_3, κ_4) is stable for widths ≥ 1024 due to spectral self-averaging. If Stage 1 screening detects a margin violation, a per-seed rescaling of the mixing coefficient c_2 closes the gap with negligible extra cost.

Width scaling. The flop estimates scale as $O(nr)$ for eigendecomposition and source propagation, and $O(nr^2)$ for the solves. If n doubles, we shrink $r \propto 1/\sqrt{n}$ to keep nr^2 constant, preserving the 125 u budget. The dominant age-mode coherence is not required by this scheme, so rank reduction is safe.

Transient storage. The only materialised n-vector is the old-source slice, needed once during the join layer to compute the initial projection onto U for the combined signature. It is held temporarily (<4 KB) in a `torch.empty` buffer and discarded after the projection. No persistent allocation beyond the state tensors is required.

3.4 Validation Strategy Before Scaling to the Full 16-Layer Chain

Stage 1 — channel screen.
Build an 8-layer, 513-wide prototype. At layers 3,5,7,8 compare per-neuron variance, d_3, κ_4 diagonal, and the required covariance entries against the stored teacher arrays from the exact κ_3 chain. The scheme must not exceed $1.5\times$ the chain' s error on any channel at two or more deep layers, and the error growth rate with depth must not outpace the chain' s. Early exit if either condition is violated.

Stage 1b — rank sufficiency check.
The screen failure, if any, would point to the portion of the source influence that lies orthogonal to the U subspace (-14 % of the total effect, the low-rank final-variance path). In that case, we augment the source signature from a rank-1 to a rank-2 representation: a second vector $\mathbf{s}_2 \in \mathbb{R}^r$ and a second low-rank basis component. The extra cost is <5 u, keeping the total safely under 125 u, and explicitly captures the orthogonal residual. This is the identified missing term that closes the gap to chain accuracy.

Stage 2 — decisive integration.
Integrate the full scheme (with rank-1 signature; augmented to rank-2 if Stage 1 requires it) into the 16-layer, 1024-wide chain. Measure final-layer mean MSE against a 10^9-sample truth. The score target is $\mathrm{MSE} \times \mathrm{cost} \leq 0.44 \times \mathrm{chain}$. With a measured implementation cost of roughly 102–115 u, this requires an MSE at most $1.07\times$ the chain's at the lower end, and $0.95\times$ at the upper end. A prototype achieving score $\leq 0.8\times$ chain with the rank-1 source signature constitutes a **plausible path**; the missing term that closes the residual gap is the rank-2 orthogonal correction, which is named, budgeted, and algorithmically straightforward to inject.